



AWS Storage Gateway

Lecture Slides

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Introduction to Storage Gateway



AWS Storage Gateway connects on-premise software applications with cloud-based storage.



File Gateway allows you run a gateway within your on-premise environment so you can interact through a SMB or NFS file-system protocol.



File Gateway to Amazon S3
Store data in S3



File Gateway to FSx for WFS
Store data in a Windows File Server



Think of: File Gateway = shared folder
Volume Gateway = remote drive



Volume Gateway

Allows you to mount s3 as local drive using the iSCSI protocol



Cached Volumes

Primary data stored on S3 and frequently access files stored locally



Stored Volumes (Non-Cached)

Primary data stored locally and entire data backed up to s3



Tape Gateway

stores files onto Virtual Library Tapes (VTLs) for backing up your files on very cost effective long term storage.

Amazon S3 File Gateway



Amazon S3 File Gateway allows your files to be stored as objects inside your S3 buckets. Access your files through a **Network File System (NFS)** or **SMB** mount point.



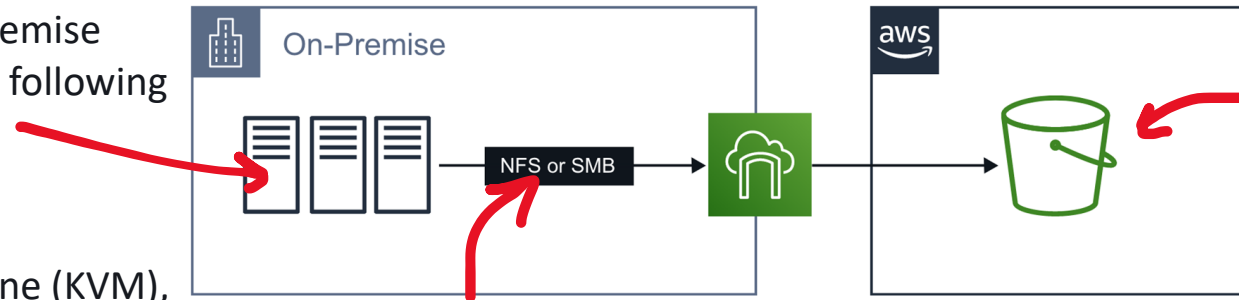
Amazon S3 provides a **cost-effective alternative** to on-premises storage

You deploy your gateway to an on premise Virtual Machine that runs one of the following hypervisors:

- VMware ESXi
- Microsoft Hyper-V
- Linux Kernel-based Virtual Machine (KVM),

The gateway can also be deployed to:

- *VMware Cloud on AWS
- AMI in Amazon EC2



For file protocols you can use:

- Network File System (NFS) Version 3 or 4.1
- Server Message Block (SMB) Version 2 or 3

Files can take full advantage of S3's offerings:

- Bucket Policies
- Versioning
- Lifecycle Management
- Cross-Region Replication
- Metadata

They act like native S3 objects.

Amazon S3 File Gateway integrates with

- IAM, KMS, CloudWatch, Cloud Trail, AWS CLI

Amazon FSx File Gateway



Amazon FSx File Gateway allows your files to be stored in Amazon FSx Windows File Storage (WFS). Allow your Windows developers to easily store data in the cloud using the tools they already know.

- you must have at least one Amazon FSx for Windows File Server file system
- You must also have on-premises access to FSx for Windows File Server
 - either through a VPN or through an AWS Direct Connect connection

You deploy your gateway to an on-premise Virtual Machine that runs one of the following hypervisors:

- VMware ESXi
- Microsoft Hyper-V
- Linux Kernel-based Virtual Machine (KVM)
- or as a hardware appliance that you order from your preferred reseller.

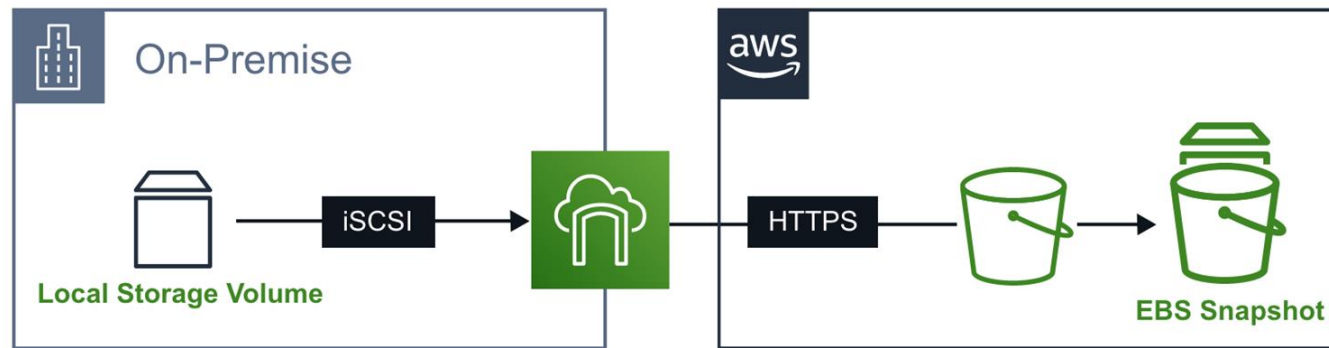
The gateway can also be deployed to:

- *VMware Cloud on AWS
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Volume Gateway



Volume Gateway presents your applications with disk volumes using the **Internet Small Computer Systems Interface (iSCSI)** block protocol.



- Data written to volumes can be asynchronously backed up as point-in-time snapshots of the volumes and stored in the cloud as AWS EBS Snapshots.
- Snapshots are incremental backups that capture only changed blocks in the volume.
- All Snapshot storage is also compressed to help minimize your storage charges.

Volume Gateway – Stored Volumes



Store Volumes store primary data locally and **asynchronously back up** that data to AWS.



- Provide your on-premises applications with low-latency access to their entire datasets while still providing durable off-site backups.
- Create storage volumes and mount them as iSCSI devices from your on-premises servers.
- Any data written to stored volumes are stored on your on-premises storage hardware.
- Amazon Elastic Block Store (EBS) snapshots are backed up to AWS S3.
- Stored Volumes can be between 1GB - 16TB in size

Hosting:

- Deployed a VM appliance
- Deployed as a Hardware appliance
- Deployed to an EC2 instance

Volume Gateway - Cached Volumes



Cache Volumes store primary data in Amazon **S3** while retaining **frequently accessed data locally** in your storage gateway.



On-Premise

**Cache Most Frequently
Accessed Blocks**



Primary Data

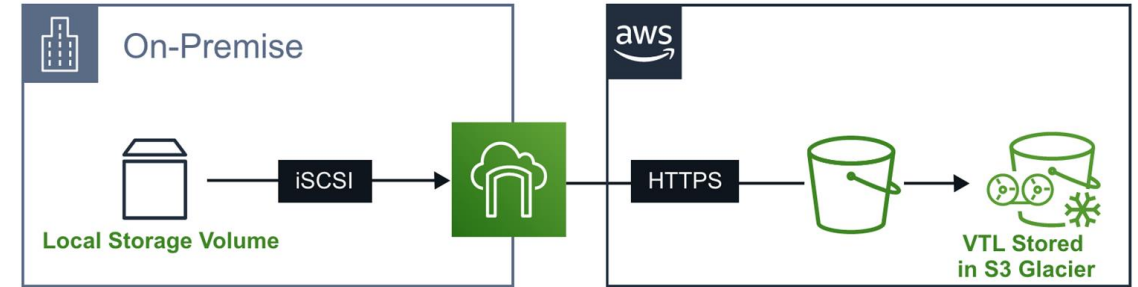
- Minimizes the need to scale your on-premises storage infrastructure while still providing your applications with low-latency data access.
 - Create storage volumes up to 32TB in size and attach them as iSCSI devices from your on-premises servers.
 - Your gateway stores data that you write to these volumes in S3 and retains recently read data in your on-premises storage gateway cache, and upload buffer storage.
 - Cached volumes can be between 1GB - 32GB in size
- Hosting:
- Deployed a VM appliance
 - VMware ESXi Hypervisor
 - Microsoft Hyper-V
 - Linux Kernel-based Virtual Machine (KVM).
 - Deployed as a Hardware appliance
 - Deployed to an EC2 instance

Tape Gateway



Tape Gateway is a durable, cost-effective solution to archive your data in the AWS Cloud

The **VTL** interface it provides lets you leverage existing tape-based backup application infrastructure.



Store data on virtual tape cartridges that you create on your tape gateway.
Tape storage has **proven readability of 30 years**

- Each tape gateway is pre-configured with a media changer and tape drives, which are available to your existing client backup applications as iSCSI devices.
- You add tape cartridges as you need to archive your data.
- Supported by NetBackup, Backup Exec, and Veeam.

Hosting:

- Deployed a VM appliance
 - VMware ESXi Hypervisor
 - Microsoft Hyper-V
 - Linux Kernel-based Virtual Machine (KVM).
- Deployed as a Hardware appliance
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Veritas Backup Exec™



Tape Gateway



A VTL media changer is analogous to a robot that moves tapes around in a physical tape library's storage slots and tape drives.

Tape Gateway has two possible medium changers

- AWS-Gateway-VTL
- **STK L700** — *emulated*



StorageTek L700 Tape Library

Different backup applications can use both or only specific media changers

- Arcserve Backup
- Bacula Enterprise V10.x
- Commvault V11
- Dell EMC NetWorker 19.5
- IBM Spectrum Protect v8.1.10
- Micro Focus (HPE) Data Protector 9 or 11.x
- Microsoft System Center 2012 R2 or 2016 Data Protection Manager
- NovaStor DataCenter/Network 6.4 or 7.1
- Quest NetVault Backup 12.4 or 13.x
- Veeam Backup & Replication 11A
- Veritas Backup Exec 2014 or 15 or 16 or 20 or 22.x
- Veritas Backup Exec 2012